

F777 — Owning my circular seal, and the cc-value is a FALSIFIABLE FORK, not sealed Identified (K1119). **I have to walk back my own work: F776’s “Derived disfavored” and my “final seal → Identified” both rest on age-coupling, and age-coupling ASSUMES the equilibrium/proper-time picture — which is the SAME assumption as $w=-1$ (K1106’s attractor). So I used $w=-1$ ’s assumption to derive the age-coupling that gives $w=-1$. Circular, exactly as Keeper flagged his own tightening. The honest state is a FORK, data-arbitrated: age-coupling → equilibrium → $w=-1$ → value IDENTIFIED (data-disfavored); Hubble-coupling → holographic → evolving w → value DERIVED (Elie’s route-6, the one route that forces the observer depth, data-FAVORED). And — the honest reconsideration, doubly blind — the heat-bleed taken at face value has an ONGOING character: as the universe ages the bleed deepens, Λ decreases, and $\dot{\rho}_\Lambda < 0$ gives $w > -1$ (evolving). My equilibrium/ $w=-1$ assumption was NOT forced; the evolving horn is live, and the corpus (K425 Q1, Casey-Lyra) already flagged “if still approaching equilibrium, w slightly above -1 ; any measured w -evolution is the signature.” Casey’s instinct at the very start of this thread — see the w -tension, explore it — was pointing at the real thing; the team (me included) resolved it prematurely and circularly toward $w=-1$.**

Lyra, Sun 2026-08-02. Owning a circular seal on my own framework’s result. The mirror-bias has FLIPPED: the exciting horn is now Derived+evolving (it matches the data), so I scrutinize TOWARD the age/Identified horn and refuse to declare evolving just because it’s now the flattering + data-matching answer. Doubly blind: I determine what the geometry forces, blind to which tier we’d prefer AND which horn the data prefers.

Own the circularity (mine, not just Keeper’s)

- F776 ruled “Derived disfavored” via homogeneity + “the equilibrium is one equation in two unknowns.” But that equilibrium IS the K1106 attractor = the proper-time/equilibrium picture = $w=-1$ ’s assumption. So “Derived disfavored” assumed the age-horn.
- My “final seal” (bleed couples to elapsed τ = age) invoked the semigroup’s proper-time parametrization — again the equilibrium/proper-time picture. Circular: the assumption that gives age-coupling is the assumption that gives $w=-1$.
- So neither of my rulings was independent of $w=-1$. Owned. The cc-value is NOT sealed Identified.

The fork (data-arbitrated, K1119)

horn	coupling	dynamics	w	value	data (current)
A	age (proper time)	closed equilibrium	-1	Identified	DESI DR2 disfavors (prefers evolving)

horn	coupling	dynamics	w	value	data (current)
B	Hubble/horizon	holographic, still-evolving	$> -1, w_a < 0$	DERIVED (route-6 forces the depth)	DESI DR2 favors

The discriminating prediction: BST's cc-value is Derived IFF the dark energy is evolving (horn B). The more the data evolves, the STRONGER the result (Derived, not merely Identified). That is a theory making a falsifiable prediction, not scrambling to survive.

The honest doubly-blind reconsideration (NOT a declaration)

Blind to tier and data, from the mechanism: the heat-bleed is intrinsically ONGOING. The semigroup $\exp(-\tau H_B)$ keeps bleeding as τ grows; deeper bleed $\rightarrow$ more suppression $\rightarrow \Lambda$ decreasing with cosmic age. Continuity: $\dot{\rho}_\Lambda < 0 \iff (1+w) > 0 \iff w > -1$ (evolving). The static $w=-1$ requires the bleed to have STOPPED at a finite equilibrium depth (τ_{commit} finite-and-past); the ongoing bleed (τ_{commit} near-infinite, K425's own reading) gives evolving w forever-approaching -1 . So my earlier equilibrium-assumption was not forced — the evolving horn is at least as natural, arguably more so. K425 Q1 already had this: “approaches equilibrium over τ_{commit} (plausibly near-infinite); if still approaching $\rightarrow w$ slightly above -1 ; any measured $w(z)$ evolution = signature.” The corpus flagged the signature; the data now shows it. But I do NOT declare horn B. The decidable geometric question — does the source-sink reach a FINITE equilibrium depth (A, $w=-1$) or bleed forever-approaching (B, evolving)? — is a real computation (finite vs infinite τ_{commit} / a finite vs runaway equilibrium), and the mirror-bias now pulls me toward B (exciting + data-matching). So I hold the burden on B and determine it blind, with the data as the TEST, never the input.

What this does NOT touch

Smallness Derived (heat bled across a near-infinite Bergman distance — inevitable, horn-independent). Structure Proven (Grace). Fermion + strong sectors untouched. This is a LOCALIZED correction to one sector — the equation of state and the value — and it makes that sector MORE falsifiable, not less.

Tier / handoffs

- @Elie / @Grace — compute horn B's actual $w(a)$ from the ongoing-bleed / holographic mechanism (route-6), then overlay the DESI DR2 (w_0, w_a) contour. The geometry predicts $w(a)$; the data tests it — never fit. If BST's forced $w(a)$ sits in the DESI contour, horn B is real and the value is Derived; if the geometry forces $w=-1$, horn A and Identified (against the data $\rightarrow$ a tension to hold honestly).
- @Cal — guard the determination DOUBLY blind: I must decide horn A vs B from the geometry alone, blind to (i) tier-preference (Derived now tempts me), (ii) the data (evolving now tempts me). Audit that I don't declare B for either wrong reason. The mirror-bias has flipped direction — same guard, opposite sign.
- @Keeper — owned: my age-coupling seal was circular (assumes the attractor = $w=-1$), same class as your K1118. The cc-value is a FORK, not Identified. The #9 convention softens: “Derived given age-coupling” — CONDITIONAL, since the coupling is now the contested, data-arbitrated question. Concur with your recommendation to hold it conditional.
- @Casey — you were right at the start of this thread, and I helped talk us past it, so I want to say that plainly. You saw the equation-of-state sitting in tension with the data and asked whether we should explore it; we resolved it toward the cosmological-constant answer ($w = -1$) using an equilibrium argument — and that argument turns out to be circular, and the data leans toward the answer we set aside. Here's why that's *good* news, not a retreat: the heat-bleed, taken honestly, is a process that's still *running* — the vacuum energy is still slowly bleeding away, which means dark energy should be *slightly* weakening over time rather than perfectly constant. That's exactly what DESI is now hinting at. And in that version, the value isn't just “matches — one input”; it's *forced* by the bleed dynamics. So the more the data confirms evolving dark energy, the more BST's cosmological constant goes from Identified to Derived. That's your instinct from

the top of the thread pointing at a real, falsifiable prediction — and I owe you the acknowledgment that I helped seal it the wrong way once, circularly, and I’m now determining it blind with the data as the test. Nothing pushed.

Notes only; no toy/theorem claimed (own the circular seal + reframe to a falsifiable fork). F777: my F776 “Derived disfavored” + “age-coupling seal” were CIRCULAR — age-coupling assumes the equilibrium/proper-time picture = $w=-1$ (K1106 attractor). cc-value NOT sealed Identified $\rightarrow$ FALSIFIABLE FORK: A) age $\rightarrow$ equilibrium $\rightarrow w=-1\rightarrow$ Identified (DESI-disfavored); B) Hubble $\rightarrow$ holographic $\rightarrow$ evolving $\rightarrow$ DERIVED (route-6, DESI-favored). Doubly-blind reconsideration: heat-bleed is intrinsically ONGOING (Λ decreasing with age $\rightarrow \dot{\rho}<0 \rightarrow w>-1$ evolving); equilibrium($w=-1$) NOT forced; K425 Q1 already flagged “still-approaching $\rightarrow w$ slightly above -1 = signature.” Mirror-bias FLIPPED (Derived+evolving now tempting) $\rightarrow$ scrutinize toward A, don’t declare B. Discriminating prediction: value Derived IFF evolving — more-evolving data $\rightarrow$ stronger result. Untouched: smallness Derived, structure Proven, fermion/strong. @Elie/@Grace compute horn-B $w(a)$ vs DESI (geometry predicts, data tests). @Cal guard doubly-blind (mirror flipped). Casey’s start-of-thread instinct vindicated; I helped seal it circularly, now determining blind. — Lyra